

1

Writing Basic SQL Statements

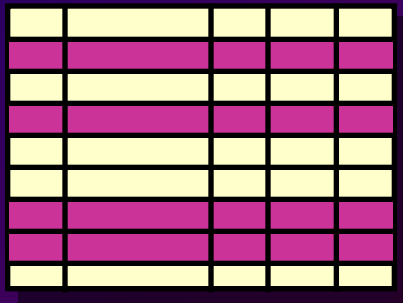
Objectives

After completing this lesson, you should be able to do the following:

- **List the capabilities of SQL SELECT statements**
- **Execute a basic SELECT statement**
- **Differentiate between SQL statements and SQL*Plus commands**

Capabilities of SQL SELECT Statements

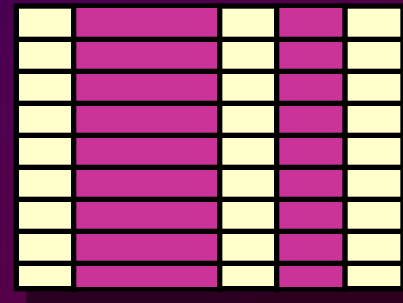
Selection



A 10x5 grid representing a table. The second, fourth, sixth, eighth, and tenth rows are highlighted in pink, illustrating the selection of specific rows from the table.

Table 1

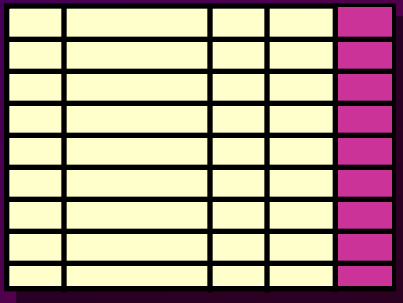
Projection



A 10x5 grid representing a table. The first, third, and fifth columns are highlighted in pink, illustrating the selection of specific columns from the table.

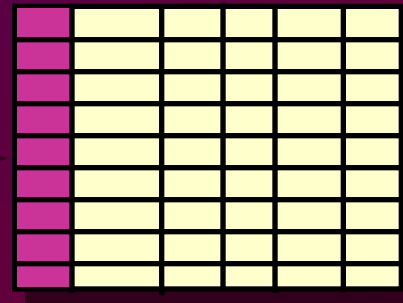
Table 1

Join



A 10x5 grid representing a table. The last column (column 5) is highlighted in pink, representing a specific attribute.

Table 1



A 10x6 grid representing a table. The first column (column 1) is highlighted in pink, representing a specific attribute.

Table 2

Basic SELECT Statement

```
SELECT    [DISTINCT] {*, column [alias], ...}  
FROM      table;
```

- SELECT identifies **what** columns.
- FROM identifies **which** table.

Writing SQL Statements

- **SQL statements are not case sensitive.**
- **SQL statements can be on one or more lines.**
- **Keywords cannot be abbreviated or split across lines.**
- **Clauses are usually placed on separate lines.**
- **Tabs and indents are used to enhance readability.**

Selecting All Columns

```
SQL> SELECT *  
2 FROM dept;
```

DEPTNO	DNAME	LOC
10	ACCOUNTING	NEW YORK
20	RESEARCH	DALLAS
30	SALES	CHICAGO
40	OPERATIONS	BOSTON

Selecting Specific Columns

```
SQL> SELECT deptno, loc  
2 FROM dept;
```

DEPTNO	LOC
10	NEW YORK
20	DALLAS
30	CHICAGO
40	BOSTON

Column Heading Defaults

- **Default justification**
 - **Left: Date and character data**
 - **Right: Numeric data**
- **Default display: Uppercase**

Arithmetic Expressions

Create expressions on NUMBER and DATE data by using arithmetic operators.

Operator	Description
+	Add
-	Subtract
*	Multiply
/	Divide

Using Arithmetic Operators

```
SQL> SELECT ename, sal, sal+300  
2 FROM emp;
```

ENAME	SAL	SAL+300
KING	5000	5300
BLAKE	2850	3150
CLARK	2450	2750
JONES	2975	3275
MARTIN	1250	1550
ALLEN	1600	1900

...

14 rows selected.

Operator Precedence

*** / + -**

- **Multiplication and division take priority over addition and subtraction.**
- **Operators of the same priority are evaluated from left to right.**
- **Parentheses are used to force prioritized evaluation and to clarify statements.**

Operator Precedence

```
SQL> SELECT  ename, sal, 12*sal+100  
2    FROM    emp;
```

ENAME	SAL	12*SAL+100
-----	-----	-----
KING	5000	60100
BLAKE	2850	34300
CLARK	2450	29500
JONES	2975	35800
MARTIN	1250	15100
ALLEN	1600	19300

...

14 rows selected.

Using Parentheses

```
SQL> SELECT ename, sal, 12*(sal+100)
2 FROM emp;
```

ENAME	SAL	12*(SAL+100)
KING	5000	61200
BLAKE	2850	35400
CLARK	2450	30600
JONES	2975	36900
MARTIN	1250	16200

...

14 rows selected.

Defining a Null Value

- A null is a value that is unavailable, unassigned, unknown, or inapplicable.
- A null is not the same as zero or a blank space.

```
SQL> SELECT  ename, job, comm  
2 FROM      emp;
```

ENAME	JOB	COMM
-----	-----	-----
KING	PRESIDENT	
BLAKE	MANAGER	
...		
TURNER	SALESMAN	0
...		

14 rows selected.

Null Values in Arithmetic Expressions

Arithmetic expressions containing a null value evaluate to null.

```
SQL> select  ename, 12*sal+comm  
2   from    emp  
3  WHERE    ename='KING' ;
```

ENAME	12*SAL+COMM
-----	-----
KING	

Defining a Column Alias

- **Renames a column heading**
- **Is useful with calculations**
- **Immediately follows column name; optional AS keyword between column name and alias**
- **Requires double quotation marks if it contains spaces or special characters or is case sensitive**

Using Column Aliases

```
SQL> SELECT  ename AS name, sal salary
2 FROM      emp;
```

NAME	SALARY
-----	-----
.	.
.	.
.	.

```
SQL> SELECT  ename "Name",
2          sal*12 "Annual Salary"
3  FROM      emp;
```

Name	Annual Salary
-----	-----
...	

Concatenation Operator

- **Concatenates columns or character strings to other columns**
- **Is represented by two vertical bars (||)**
- **Creates a resultant column that is a character expression**

Using the Concatenation Operator

```
SQL> SELECT  ename||job AS "Employees"  
2  FROM      emp;
```

Employees

KINGPRESIDENT

BLAKEMANAGER

CLARKMANAGER

JONESMANAGER

MARTINSALESMAN

ALLENSALESMAN

...

14 rows selected.

Literal Character Strings

- A literal is a character, expression, or number included in the **SELECT** list.
- Date and character literal values must be enclosed within single quotation marks.
- Each character string is output once for each row returned.

Using Literal Character Strings

```
SQL> SELECT ename || ' ' || 'is a' || ' ' || job  
2          AS "Employee Details"  
3 FROM      emp;
```

```
Employee Details  
-----  
KING is a PRESIDENT  
BLAKE is a MANAGER  
CLARK is a MANAGER  
JONES is a MANAGER  
MARTIN is a SALESMAN  
...  
14 rows selected.
```

Duplicate Rows

The default display of queries is all rows, including duplicate rows.

```
SQL> SELECT deptno  
2 FROM emp;
```

```
DEPTNO  
-----  
10  
30  
10  
20  
...  
14 rows selected.
```

Eliminating Duplicate Rows

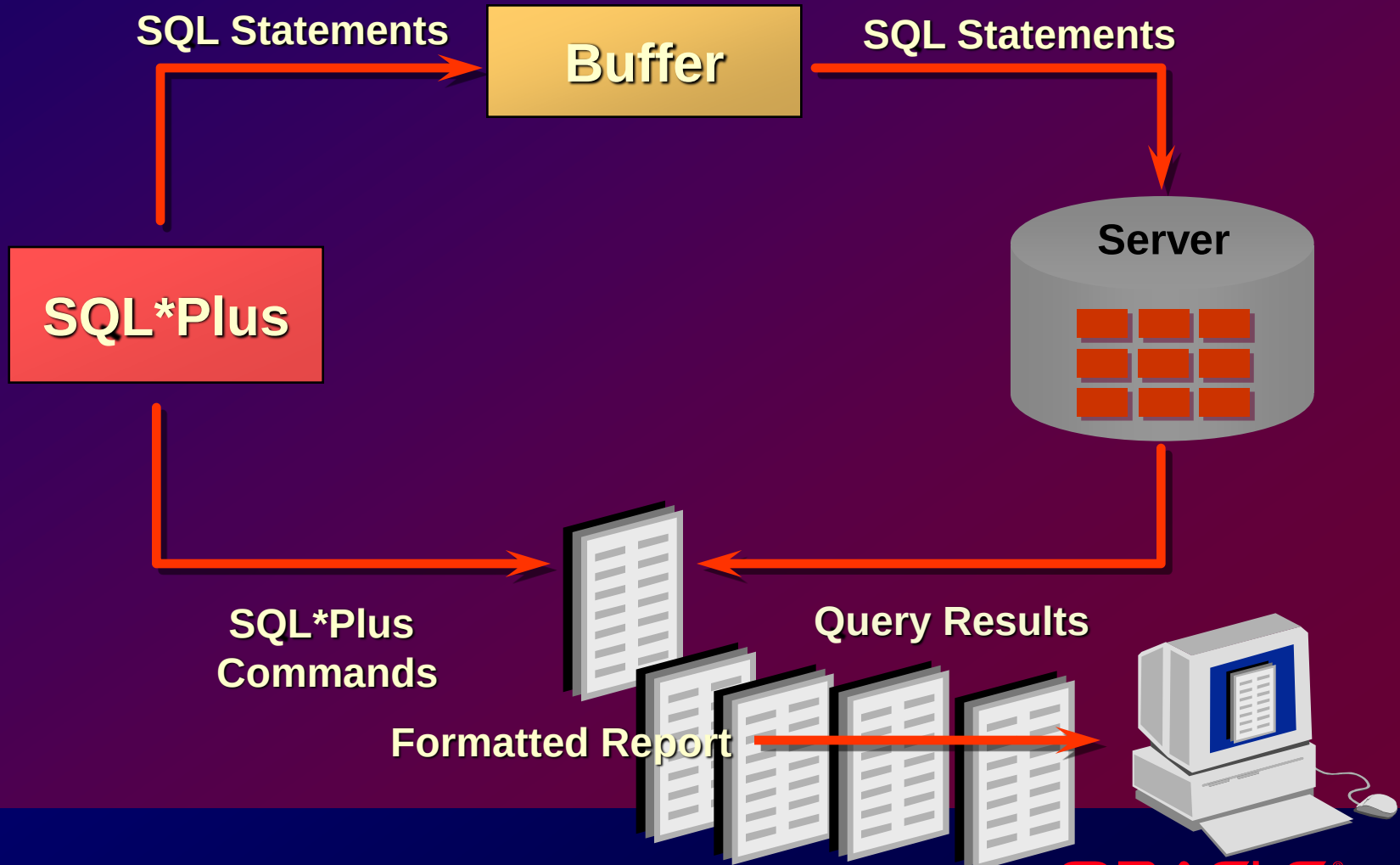
Eliminate duplicate rows by using the **DISTINCT** keyword in the **SELECT** clause.

```
SQL> SELECT DISTINCT deptno  
2 FROM emp;
```

DEPTNO

10
20
30

SQL and SQL*Plus Interaction



SQL Statements Versus SQL*Plus Commands

SQL

- A language
- ANSI standard
- Keyword cannot be abbreviated
- Statements manipulate data and table definitions in the database



SQL*Plus

- An environment
- Oracle proprietary
- Keywords can be abbreviated
- Commands do not allow manipulation of values in the database

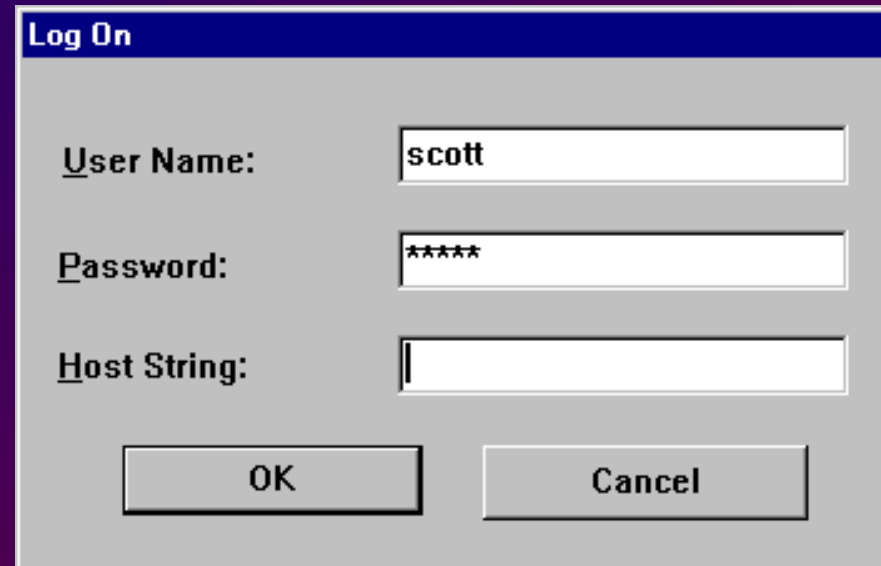


Overview of SQL*Plus

- **Log in to SQL*Plus.**
- **Describe the table structure.**
- **Edit your SQL statement.**
- **Execute SQL from SQL*Plus.**
- **Save SQL statements to files and append SQL statements to files.**
- **Execute saved files.**
- **Load commands from file to buffer to edit.**

Logging In to SQL*Plus

- From Windows environment:



A screenshot of a Windows-style dialog box titled "Log On". It contains three input fields: "User Name:" with the text "scott", "Password:" with masked characters "*****", and "Host String:" which is empty. At the bottom are "OK" and "Cancel" buttons.

- From command line:

```
sqlplus [username[/password  
[@database]]]
```

Displaying Table Structure

Use the SQL*Plus DESCRIBE command to display the structure of a table.

```
DESC[RIBE] tablename
```

Displaying Table Structure

```
SQL> DESCRIBE dept
```

Name	Null?	Type
-----	-----	-----
DEPTNO	NOT NULL	NUMBER(2)
DNAME		VARCHAR2(14)
LOC		VARCHAR2(13)

SQL*Plus Editing Commands

- **A[PPEND] *text***
- **C[HANGE] / *old* / *new***
- **C[HANGE] / *text* /**
- **CL[EAR] BUFF[ER]**
- **DEL**
- **DEL *n***
- **DEL *m n***

SQL*Plus Editing Commands

- I[INPUT]
- I[INPUT] *text*
- L[IST]
- L[IST] *n*
- L[IST] *m n*
- R[UN]
- *n*
- *n text*
- *0 text*

SQL*Plus File Commands

- **SAVE *filename***
- **GET *filename***
- **START *filename***
- **@ *filename***
- **EDIT *filename***
- **SPOOL *filename***

Summary

```
SELECT    [DISTINCT] {*,column [alias],...}  
FROM      table;
```

Use SQL*Plus as an environment to:

- **Execute SQL statements**
- **Edit SQL statements**

Practice Overview

- **Selecting all data from different tables**
- **Describing the structure of tables**
- **Performing arithmetic calculations and specifying column names**
- **Using SQL*Plus editor**